

Variational Cost for the Burgers Equation

Construction of the cost function and its quantum evaluation

1. From the PDE to a variational target

For the viscous Burgers equation

$$\partial_t u + u \partial_x u = \nu \partial_x^2 u,$$

the spatial derivatives are first discretized. Using centered finite differences,

$$D_1 u_j = \frac{u_{j+1} - u_{j-1}}{2\Delta x}, \quad D_2 u_j = \frac{u_{j+1} + u_{j-1} - 2u_j}{\Delta x^2}.$$

One explicit-Euler step gives the target vector

$$\boxed{|u_{\text{tar}}\rangle = |u_n\rangle + \Delta t (\nu D_2 |u_n\rangle - u_n D_1 |u_n\rangle).} \quad (1)$$

This target need not lie exactly on the manifold of states representable by the chosen parameterized quantum circuit. The variational task is therefore to find the representable vector closest to it.

The code separates the overall amplitude from the normalized circuit state,

$$|u(\theta)\rangle = a |\psi(\phi)\rangle, \quad \langle \psi | \psi \rangle = 1,$$

where the first variational parameter is the amplitude a and the remaining parameters ϕ specify the quantum circuit. At the next time step,

$$|u_{n+1}\rangle = a_{n+1} |\psi_{n+1}\rangle.$$

2. Constructing the cost function

The natural least-squares objective is

$$L(\theta_{n+1}) = \|a_{n+1} |\psi_{n+1}\rangle - |u_{\text{tar}}\rangle\|^2. \quad (2)$$

Expanding,

$$L = |a_{n+1}|^2 - 2 \text{Re}[a_{n+1}^* \langle \psi_{n+1} | u_{\text{tar}} \rangle] + \langle u_{\text{tar}} | u_{\text{tar}} \rangle. \quad (3)$$

During the optimization of the *new* parameters, $|u_{\text{tar}}\rangle$ is fixed because it is constructed only from the previous step. Hence $\langle u_{\text{tar}} | u_{\text{tar}} \rangle$ is a constant and can be dropped without changing the minimizer. The implemented cost is therefore

$$\boxed{C(\theta_{n+1}) = |a_{n+1}|^2 - 2 \text{Re}[a_{n+1}^* \Omega], \quad \Omega \equiv \langle \psi_{n+1} | u_{\text{tar}} \rangle.} \quad (4)$$

Thus minimizing C is equivalent to projecting the explicit-Euler target onto the variational manifold.

3. How the overlap Ω is assembled

Write $|u_n\rangle = a_n|\psi_n\rangle$ and introduce the periodic shift operator S . Since

$$D_2 = \frac{S + S^{-1} - 2I}{\Delta x^2},$$

the linear part of Eq. (1) produces

$$\Omega_{\text{const}} = a_n \left(1 - \frac{2\nu\Delta t}{\Delta x^2} \right) \langle \psi_{n+1} | \psi_n \rangle, \quad (5)$$

$$\Omega_{\text{diff}} = a_n \frac{\nu\Delta t}{\Delta x^2} \left(\langle \psi_{n+1} | S | \psi_n \rangle + \langle \psi_{n+1} | S^{-1} | \psi_n \rangle \right). \quad (6)$$

For the nonlinear convection term, the centered derivative gives a forward-minus-backward structure,

$$\Omega_{\text{nl}} = -\frac{\Delta t}{2\Delta x} a_n^2 (\mathcal{N}_+ - \mathcal{N}_-), \quad (7)$$

where $\mathcal{N}_{\pm}$ denote the nonlinear overlaps generated by the forward/backward shifted products. Therefore

$$\boxed{\Omega = \Omega_{\text{const}} + \Omega_{\text{diff}} + \Omega_{\text{nl}}.} \quad (8)$$

This is exactly the decomposition used by **burgers_overlap**: constant contribution, symmetric diffusion contribution, and nonlinear convection contribution.

4. How the cost is evaluated on a quantum computer

The important point is that one does *not* need to reconstruct the entire vector u_{tar} by tomography. Instead, the optimizer only needs the scalar quantities appearing in C . For a trial parameter set θ_{n+1} :

1. Prepare the old variational state $|\psi_n\rangle$ and the trial new state $|\psi_{n+1}\rangle$ using parameterized quantum circuits.
2. Estimate the ordinary overlap

$$\langle \psi_{n+1} | \psi_n \rangle$$

and the shifted overlaps

$$\langle \psi_{n+1} | S | \psi_n \rangle, \quad \langle \psi_{n+1} | S^{-1} | \psi_n \rangle.$$

In the supplied implementation these quantities are passed through **hadamard_test**. On quantum hardware, a Hadamard-test-type ancilla circuit can estimate the real and, when needed, imaginary parts of these overlaps from repeated measurements.

3. Estimate the two nonlinear quantities $\mathcal{N}_+$ and $\mathcal{N}_-$. In the supplied code this is delegated to **nonlinear_overlap**. These measurements encode the quadratic product $u_n D_1 u_n$ without first classically constructing the whole updated vector.
4. Combine the measured numbers classically to form

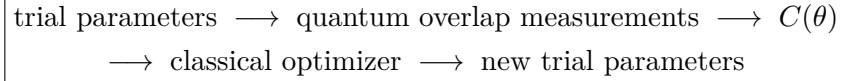
$$\Omega = \Omega_{\text{const}} + \Omega_{\text{diff}} + \Omega_{\text{nl}},$$

and then evaluate

$$C = |a_{n+1}|^2 - 2 \operatorname{Re}(a_{n+1}^* \Omega).$$

5. A classical optimizer changes a_{n+1} and the circuit parameters ϕ_{n+1} and repeats the quantum measurements until the cost is minimized.

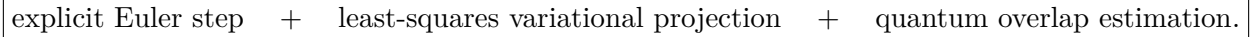
Hence the quantum-classical loop is



The notebook uses `get_expval_ideal` for ideal expectation values and COBYLA for this outer optimization. A shot-based implementation can instead estimate the same expectation values statistically.

5. Interpretation

The method is best summarized as



It is therefore different from a continuous-time McLachlan/TDVP method, where one derives differential equations directly for $\dot{\theta}$. Here, each discrete Euler target is constructed implicitly through measurable overlaps, and the next variational state is obtained by minimizing its distance from that target.